

Supplemental Online Content

Martin DS, Gould DW, Shahid T, the UK-ROX Investigators. Conservative oxygen therapy in mechanically ventilated critically ill adult patients: the UK-ROX randomized clinical trial. *JAMA*. Published online June 12, 2025. doi:10.1001/jama.2025.9663

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This supplemental material has been provided by the authors to give readers additional information about their work.

eMethods

Treatment exposure

Exposure to the intervention was assessed in patients with enhanced data collection. The following parameters were calculated for each treatment group to assess separation in exposure to oxygen:

- SpO_2 , FiO_2 , SaO_2 and PaO_2 – mean (SD) and median (IQR) of the median value for each patient, overall and while receiving oxygen, and difference in means between treatment groups with 95% confidence interval
- Total time spent: on room air with recorded $\text{SpO}_2 > 92\%$; receiving oxygen with recorded $\text{SpO}_2 > 92\%$; within target range (88-92%); and below 88% – mean (SD), median (IQR) and difference in means between treatment groups with 95% confidence interval.
- The total dose of O_2 that each patient received, calculated in 100%-equivalent hours (a measure of the cumulative exposure to excess oxygen, whereby time spent on room air represents zero additional O_2 , and one hour on 100% O_2 or two hours on 60.5% O_2 would both represent one 100%-equivalent hour of O_2) – mean (SD), median (IQR) and both absolute and percentage difference in means between treatment groups with 95% confidence interval.

Treatment adherence

Treatment adherence was recorded over the first 10 days of treatment or until refusal of consent, ICU discharge or death (whichever came first). In the conservative oxygen therapy group, where a patient was receiving supplemental oxygen (i.e. $\text{FiO}_2 > 0.21$) and SpO_2 was above 92%, a failure to reduce FiO_2 for at least 3 consecutive hours would be defined as a potential protocol deviation. Potential protocol deviations triggered a query to the participating site who were asked to provide a justification. Potential protocol deviations, and site responses to queries, were reviewed by the Trial Management Group to determine whether or not they were true deviations. Examples of 'acceptable' reasons for deviating from the treatment protocol included temporary increases in FiO_2 in response to acute desaturation or adverse event(s); targeting SaO_2 due to poor SpO_2 trace; and changes in clinical status of the patient to palliative care. As the comparator was usual care, there was no assessment of treatment adherence in that group.

Consent procedures

Patients eligible for the UK-ROX trial become so during a period of critical illness.

Mechanical ventilation is an invasive procedure initiated as a life-saving measure, during an emergency clinical situation. It requires the use of sedative and analgesic drugs, meaning most patients will lack mental capacity to provide consent. Additionally, the urgent and life-threatening nature of the clinical situation can cause profound distress for relatives (including the patient's personal consultee). This raises ethical concerns around a relative's ability to process complex trial information at a time of great distress and to decide whether the patient should be enrolled into the trial. For these reasons, attempts to obtain either prior informed consent from the patient, or the prior opinion of a personal consultee, were deemed to be inappropriate.

The UK-ROX trial therefore adopted a research without prior consent (RWPC) model, also referred to as deferred consent, whereby eligible patients were randomized to receive the assigned treatment as soon as possible (and no later than 12 hours after fulfilling the eligibility criteria). This is an accepted consent model in adult emergency and critical care research where participants lack mental capacity and minimizes the distress and additional burden on families during a distressing time. This consent model is covered by an Emergency Waiver of Consent under the Mental Capacity Act and was approved by South Central - Oxford C Research Ethics Committee (reference: 20/SC/0423).

In the very rare situation where a patient was deemed by the treating clinical team to have full mental capacity and was able to give informed consent at the point of meeting the trial's eligibility criteria, they were approached directly prior to randomization for verbal or other non-written consent to take part in the trial. If they were able to provide verbal or other non-written consent, they were then followed up for full written informed consent at a later date. If the participant gave prospective verbal or non-written consent then subsequently lost mental capacity, the opinion of a personal or nominated consultee was sought to advise on their continuation in the trial.

In most cases, it was not possible to involve trial participants in the consenting process early on. Instead, consent was obtained from patients once they had stabilized and were deemed to have regained capacity. In the interim, once notified of the enrolment of a patient into the UK-ROX trial, a delegated member of the site research team approached the participant's personal consultee (in person or via telephone) as soon as was deemed to be appropriate and practically possible. The trial was discussed, and their opinion was sought about the patient's likely wishes and feelings regarding participating in the trial. The aim was that this approach would take place within 24-48 hours of randomization, but allowances were made

to extend this until the patient's medical situation was no longer an emergency. When approached in person, the personal consultee was provided with a personal consultee information sheet, containing the information provided on the patient information sheet, supplemented with information on why the personal consultee was approached at this stage. Personal consultees were given time to read the personal consultee information sheet and have an opportunity to ask any questions they may have about the patients' participation in the UK-ROX trial. A personal consultee opinion form was then provided for the personal consultee to sign. If the personal consultee advised that, in their opinion, the patient would not choose to participate in the trial, then the trial treatment was stopped and the personal consultee asked whether, in their opinion, the patient would be willing to continue with ongoing data collection.

In the rare situation where no personal consultee was available (or one was available, but did not wish to be consulted) a nominated consultee was approached for their opinion instead. The nominated consultee could include an Independent Mental Capacity Advocate appointed by the hospital or an independent doctor (i.e. not involved in the trial). The opinion of the nominated consultee was sought in the same manner as for the personal consultee. Additionally, in the situation where the patient had died before a personal consultee could be approached, a nominated consultee was appointed instead.

Upon the participant's recovery and regaining of mental capacity, the patient was approached directly. The patient was provided with the patient information sheet. The patient's decision was final and superseded that of the personal or nominated consultee, where there was any disagreement.

In the situation where the patient was discharged from hospital with mental capacity prior to confirming their consent decision, an experienced member of the site research team attempted to contact the patient by telephone within 14 working days of ultimate hospital discharge. They informed them of their involvement in the UK-ROX trial; provided information about the trial; and sought their consent. Where appropriate, the patient information sheet was sent to the patient by email or by post. The outcome of the telephone call was then documented and signed by person seeking consent on a telephone consent form.

If a patient declined informed deferred consent, or a consultee advised that they believe the patient would not choose to participate in the trial, and, if a patient or their consultee (personal or nominated) withdrew consent/opinion at any time during the trial, that decision was respected and abided by. All data up to the point of this decision was retained in the trial, unless the patient or consultee requested otherwise.

For patients who declined informed deferred consent, or a consultee advised that they believe the patient would not choose to participate in the trial, but did not request removal of all data, the site entered minimal pseudonymized data required for the primary outcome [survival status (alive/dead) at 90 days after randomization] only. No patient identifiable data were recorded and no further contact with the patient/consultee about the trial was required. As patient identifiers were not collected, secondary outcomes were not available for these patients.

Data collection

To maximise efficiency, trial specific patient data collection was linked with routine data from the Case Mix Programme (CMP) national clinical audit and Civil Registrations of Death dataset. Trial specific data were entered onto a secure electronic case report form database and with validation checks. Trial specific data collection for all patients included randomisation data; information related to consent processes, patient identifiers to enable linkage to routine datasets. For participants selected for enhanced data collection, trial specific data collection also included hourly SpO₂, FiO₂ and arterial blood gas values.

The CMP is the national clinical audit for adult ICUs in England, Wales and Northern Ireland. Data are collected by participating ICUs according to strict definitions, covering: patient characteristics (age, sex, ethnicity, and admission source); clinical information on admission to ICU (primary reason for admission, comorbidities and physiology during first 24 hours of admission required for illness severity scoring); interventions received in ICU (organ support according to the UK Department of Health Critical Care Minimum Dataset); and outcome (ICU and hospital mortality).¹

The Civil Registrations of Deaths Dataset is held by NHS England and includes details of all registered deaths in England and Wales.²

Outcomes

Primary clinical outcome

90-day mortality

The primary clinical outcome was 90-day mortality, defined as death due to any cause by 90 days following randomization.

Date of death was collected from linked death registration records, the CMP or directly from the sites. Where data were discrepant between sources, quality assurance checks were conducted with sites and the site data considered paramount. Where sites were unable to validate discrepancies in linked records, deaths identified by either linked data source were considered true, with death registrations considered paramount for discrepant dates of death.

Secondary clinical outcomes

Mortality at discharge from the ICU and acute hospital

Patient outcomes (alive/dead) at discharge from ICU and acute hospital were recorded as part of routine CMP data collection and were collected from linked CMP records, supplemented by data collected directly from sites where necessary. Mortality at discharge from the ICU was defined as death due to any cause before discharge to any location providing a level of care less than Level 2 (high dependency care). Mortality at discharge from acute hospital was defined as death due to any cause before discharge from acute hospital. Patients transferred from the original acute hospital to another acute hospital were followed up until they had left acute hospital. ICU and hospital mortality were censored at 90 days following randomization.

Mortality at 60 days and one year

Mortality at 60 days and one year was defined as death due to any cause by 60 days and one year respectively following randomization and otherwise derived as per mortality at 90 days. As the trial ended when the final patient reached 90 days, some patients did not reach one year follow-up and were excluded from analysis of this outcome.

Duration of ICU and acute hospital stay

Duration of ICU and acute hospital stay were derived from linked CMP records. Duration of ICU stay was calculated as the sum of the duration (in linear time) from the date and time of randomization to the date and time of first discharge from the critical care unit or death in the ICU plus the duration of any subsequent admissions to the ICU within the same acute hospital stay.

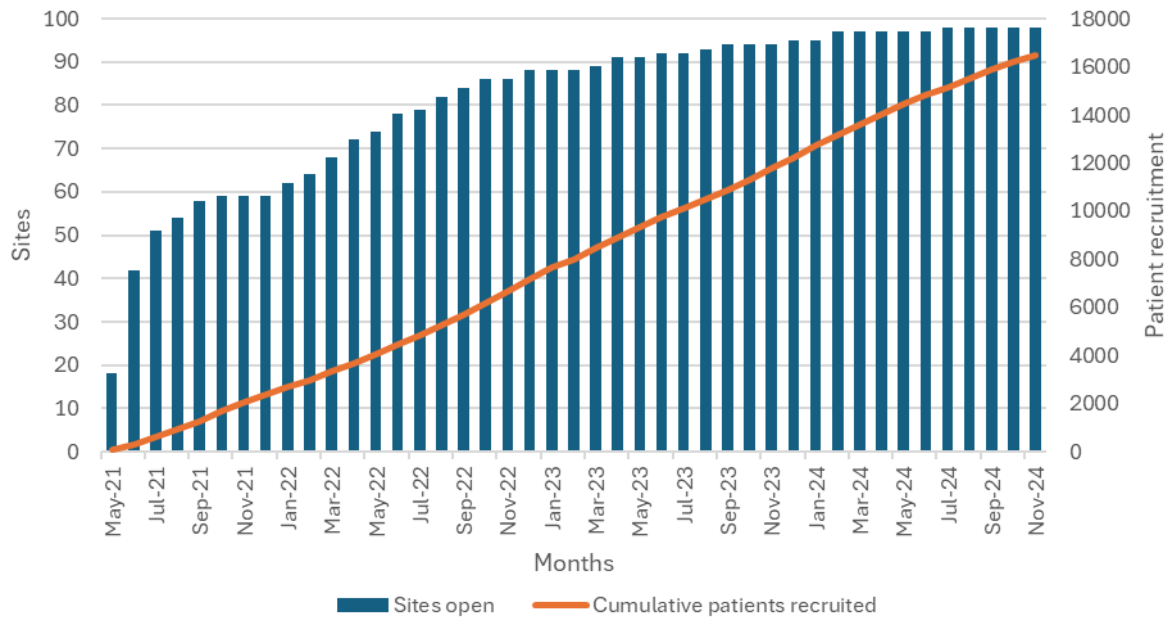
Duration of acute hospital stay was calculated as the duration in calendar days (as times of hospital admission and discharge were not available) from the date of randomization to the date of acute hospital discharge or death in acute hospital.

Days alive and free from organ support at 30 days

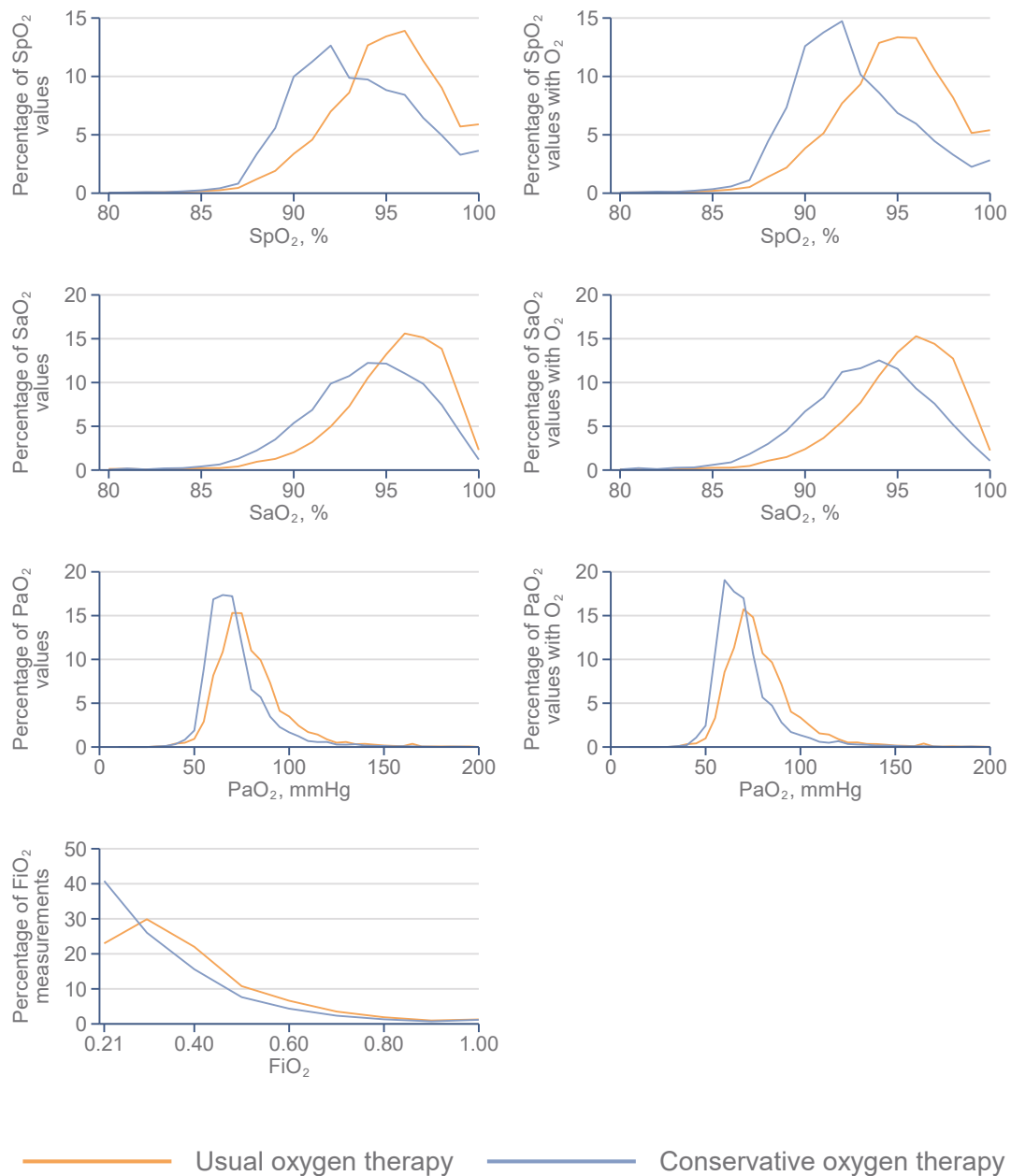
Days alive and free from organ support at 30 days is an ordinal composite outcome with death by day 30 being ranked as the worst outcome. Organ support was defined as receipt of respiratory, cardiovascular or renal support during any ICU stay within the first 30 days following randomization, according to the UK Department of Health Critical Care Minimum Dataset. Patients who died on or before 30 days following randomization were assigned the worst possible score of -1. Surviving patients were ranked according to the number of calendar days (00:00 to 23:59) on which any respiratory, cardiovascular or renal support was received at any time during that day, starting from day 1 (the day on which the patient was randomized) up to and including day 30. Any days outside of an ICU were assumed to be free of organ support. For patients recruited when the site was collecting data to Version 3.1 of the CMP Dataset Specification (when only total calendar days of organ support were collected, and not organ support on each day), all organ support received was assumed to be at the start of the critical care stay and therefore the total days of organ support was assumed to be the maximum of the days of respiratory, cardiovascular and renal support reported, up to 30 days.

Supplemental results

eFigures

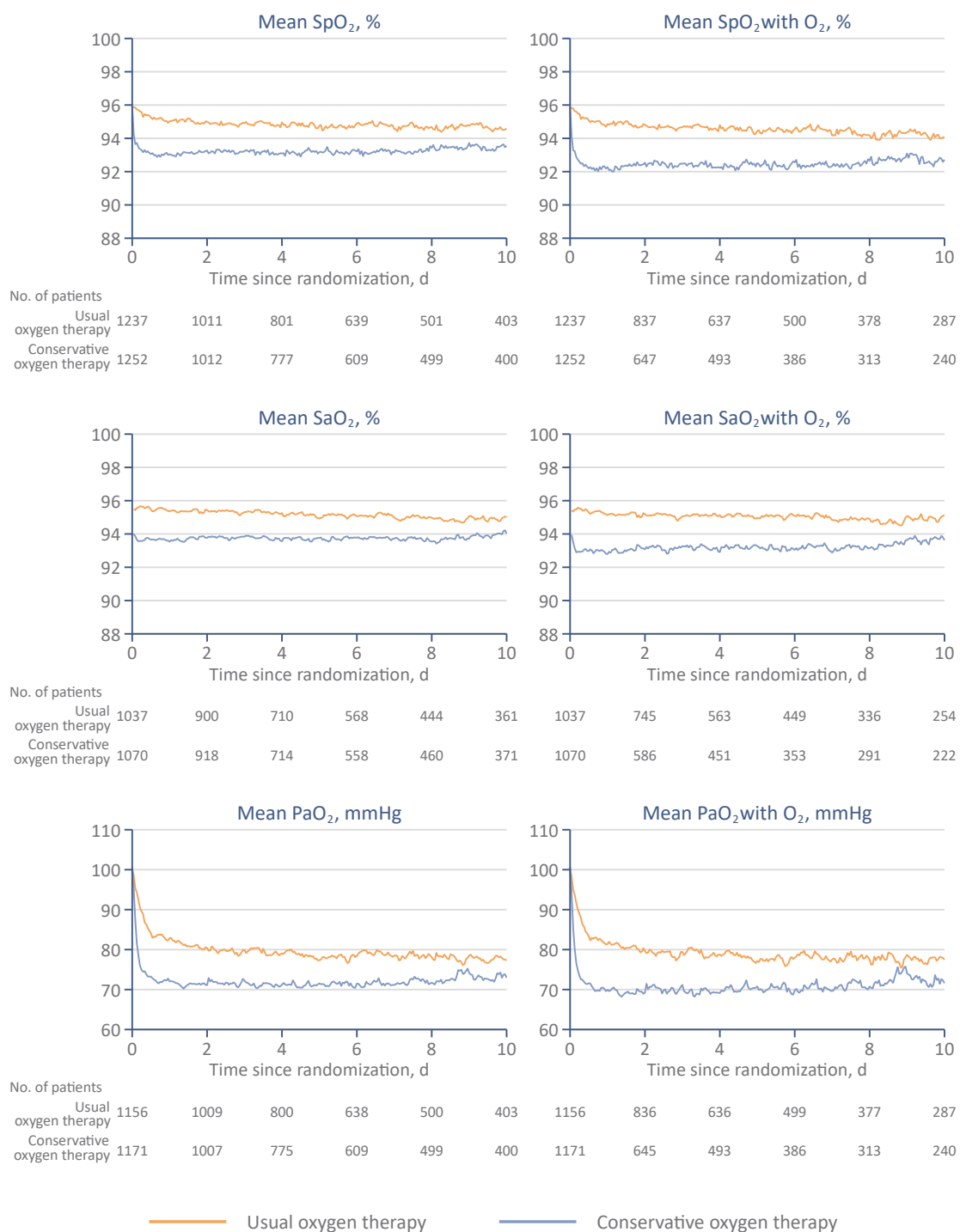


eFigure 1: Site and patient recruitment



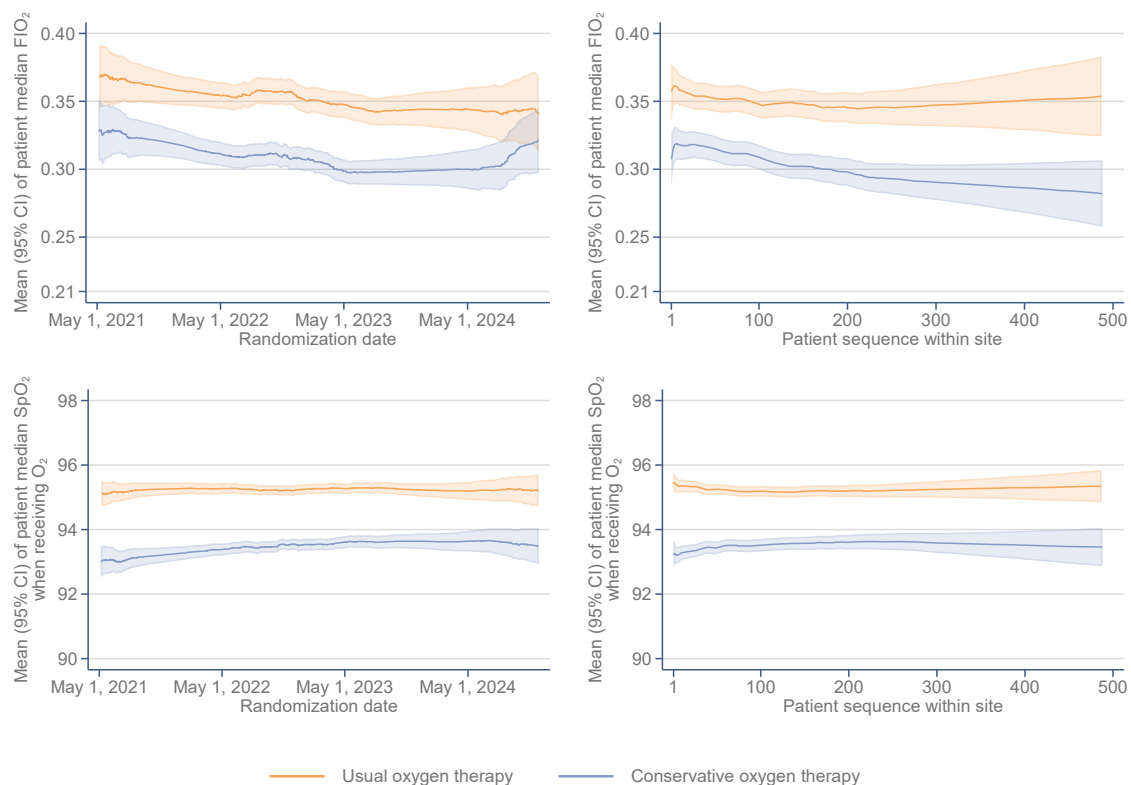
eFigure 2: Frequency distributions of SpO₂, SaO₂, PaO₂ and FiO₂ values among patients with enhanced data collection (N=2489)

SI conversion factors: To convert PaO₂ to kPa, multiply values by 0.133.



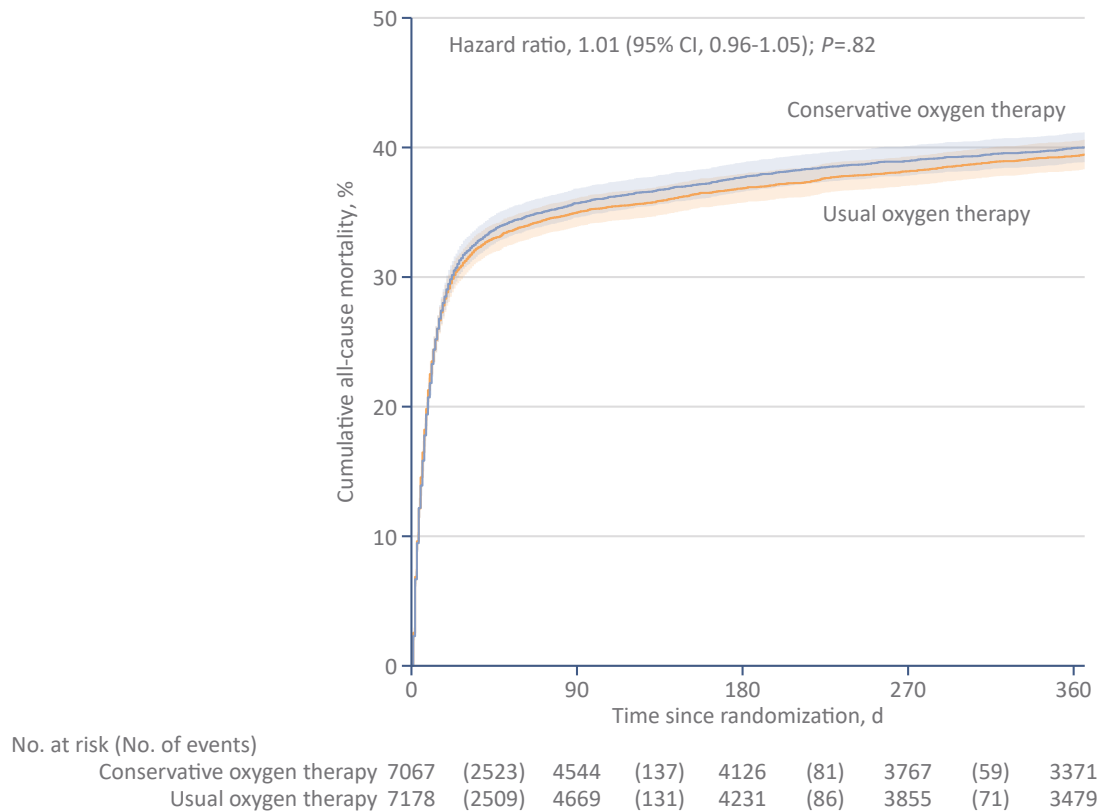
eFigure 3. Mean SpO₂, SaO₂ and PaO₂ over the first 10 days following randomization for patients with enhanced data collection (N=2489)

SI conversion factors: To convert PaO₂ to kPa, multiply values by 0.133.



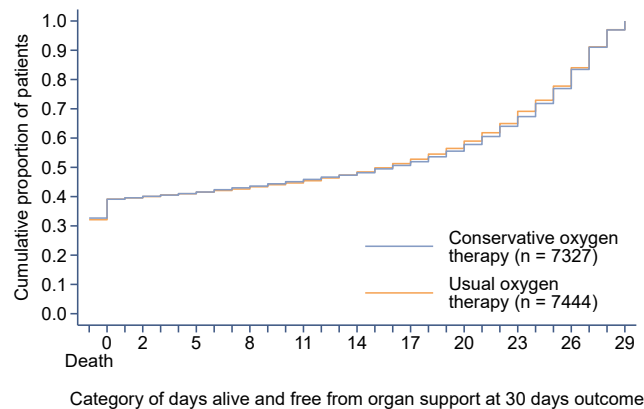
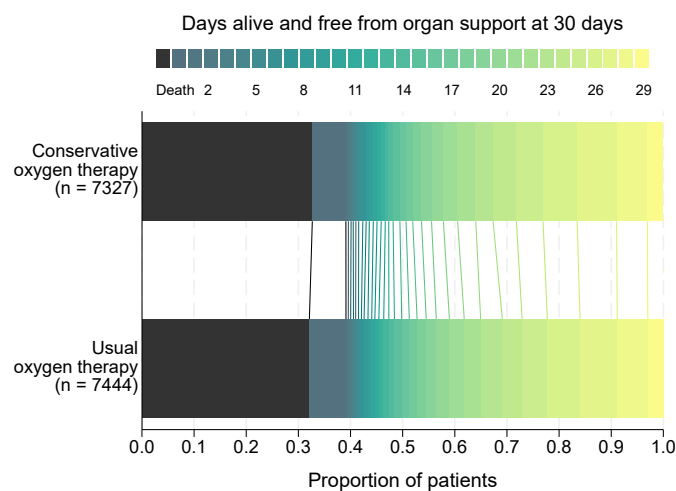
eFigure 4. Separation in FIO_2 and SpO_2 when receiving oxygen over time and by patient sequence within site for patients with enhanced data collection (N=2489)

Running line smoother of the mean value of patient median measurements over the first 10 days following randomization.



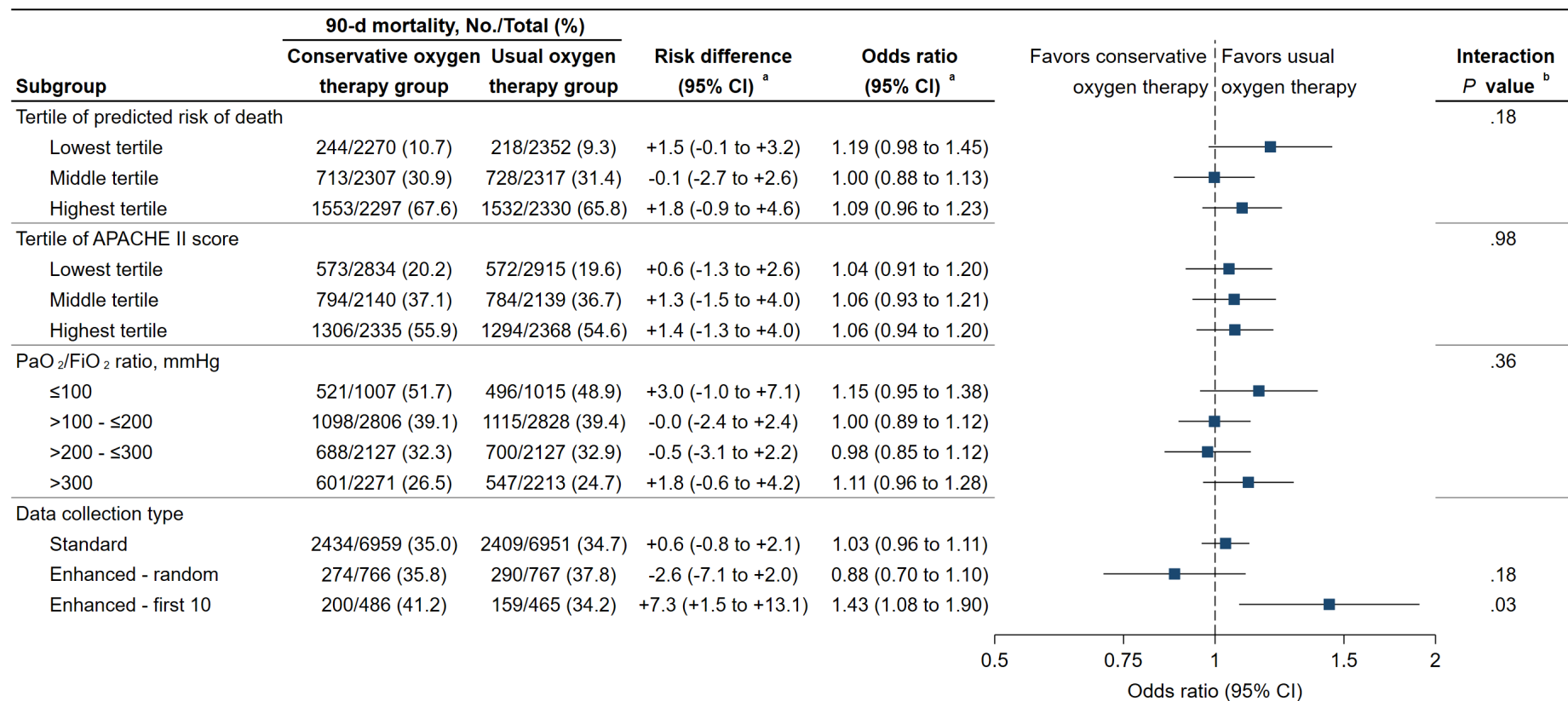
eFigure 5. Kaplan-Meier plot of cumulative all-cause mortality to 1 y following randomization

All randomized patients were included when calculating hazard ratio, excluding 66 patients who did not consent to the trial and who requested removal of all data (28 conservative oxygen therapy, 38 usual oxygen therapy). Surviving patients were right-censored at last known date alive (where trial consent was obtained), at 90 days (where trial consent was not obtained and only survival status at 90 days was recorded) or at date of withdrawal of consent. Patients known to have died within 90 days but where date of death was unavailable having declined/withdrawn consent to further follow-up and data linkage ($n=342$, 2.1%) were interval censored between the date of withdrawal and day 90 (where the patient declined/withdrew) or left-censored at day 90 (where the patient's consultee declined/withdrew). Only patients with consent for data linkage are included in the Kaplan-Meier plot. The median (IQR) follow-up time (using the reverse Kaplan Meier method) was 365 days (277 to 365) for the conservative oxygen therapy group and 365 days (292 to 365) for the usual oxygen therapy group.

A**B**

eFigure 6. Distribution of days alive and free from organ support at 30 days

Panel A shows the cumulative proportion of patients in each treatment group with each value of number of days alive and free from organ support during the first 30 days following randomization, with death listed first on the axis corresponding to a value of -1. Curves that rise more steeply indicate a less favorable distribution in the number of days alive and free from organ support. The height of the curve at the start, at -1, indicates the mortality on or before day 30 in each group. The height of the curve at any point from 0 to 29 days indicates the proportion of patients with that number of days alive and free from organ support, or fewer, or who had died by day 30 (e.g., the height at day 20 indicates the proportion of patients that survived to 30 days with ≤20 days free from organ support or who had died by day 30). Panel B shows the distribution of days alive and free from organ support as horizontally stacked proportions for each treatment group. Blue represents worse outcomes and yellow represents better outcomes.



eFigure 7. Post-hoc subgroup analyses

CI, confidence interval.

^a Adjusted for site, diagnostic subgroup, age, SpO₂, PaO₂/FiO₂ ratio, confirmed/highly suspected COVID-19, and date of randomization

^b *P* value for test of interactions in the odds ratio in adjusted multilevel logistic regression model

eTable 1. Patients randomized by site

Hospital Name	Patients Randomized
Sunderland Royal Hospital	863
St George's Hospital	626
Bristol Royal Infirmary	482
Grange University Hospital	473
St James's University Hospital	442
Worthing Hospital	429
Norfolk and Norwich University Hospital	411
Tunbridge Wells Hospital	384
Royal Bournemouth Hospital	349
Derriford Hospital	348
University Hospital Southampton	348
Royal Blackburn Teaching Hospital	325
Kettering General Hospital	318
Royal Cornwall Hospital	318
Queen Alexandra Hospital	312
Leeds General Infirmary	302
Pinderfields Hospital	299
St Richard's Hospital	279
Queen Elizabeth Hospital, Woolwich	277
Royal Oldham Hospital	276
Royal Victoria Infirmary	265
University Hospital Coventry	244
Aintree University Hospital	236
Stoke Mandeville Hospital	230
Poole Hospital	225
Royal Stoke University Hospital	222
Torbay Hospital	219
University Hospital Lewisham	216
Warrington Hospital	213
Maidstone Hospital	206
Tameside General Hospital	185
Darlington Memorial Hospital	180
Glangwili General Hospital	179
West Middlesex University Hospital	176
Bradford Royal Infirmary	176
Broomfield Hospital	173
Ulster Hospital	168
Leicester Royal Infirmary	167
Chesterfield Royal Hospital	167
Royal Devon and Exeter Hospital, Wonford	165
York Hospital	151
Blackpool Victoria Hospital	144
Royal Free Hospital	137
Rotherham Hospital	137
Sandwell General Hospital	136
Queen's Hospital, Burton	132
Southmead Hospital	130
Royal Papworth Hospital	130
Royal United Hospital, Bath	128
Hospital Name	Patients Randomized
Salford Royal Hospital	127
Gloucestershire Royal Hospital	126
Northern General Hospital	124
Worcestershire Royal Hospital	121

Whiston Hospital	121
Countess of Chester Hospital	119
Northwick Park Hospital	118
Medway Maritime Hospital	116
Musgrove Park Hospital	111
Craigavon Area Hospital	111
Warwick Hospital	109
James Cook University Hospital	104
King's Mill Hospital	102
Queen's Medical Centre	100
Northampton General Hospital	100
Scarborough General Hospital	100
King's College Hospital	97
Dorset County Hospital	96
Glan Clwyd Hospital	96
Ealing Hospital	95
Liverpool Heart and Chest Hospital	94
Chelsea and Westminster Hospital	93
Royal Sussex County Hospital	83
Colchester General Hospital	79
Queen Elizabeth The Queen Mother Hospital	75
Birmingham City Hospital	72
William Harvey Hospital	69
Hull Royal Infirmary	68
Royal Liverpool University Hospital	66
Fairfield General Hospital	54
Addenbrooke's Hospital	51
North Middlesex University Hospital	50
Queen's Hospital, Romford	49
Princess Royal Hospital, Haywards Heath	49
Alexandra Hospital, Redditch	45
Royal Bolton Hospital	40
Freeman Hospital	29
Harrogate District Hospital	28
Princess Royal University Hospital, Farnborough	25
Wrexham Maelor Hospital	19
Russells Hall Hospital	15
Royal Surrey County Hospital	14
Morriston Hospital	10
Queen Elizabeth Hospital, King's Lynn	9
Altnagelvin Hospital	8
Queen Elizabeth Hospital, Gateshead	7
Southend University Hospital	7
The Walton Centre	1
Total	16 500

eTable 2. Additional patient characteristics

	Conservative oxygen therapy group (n = 8230)	Usual oxygen therapy group (n = 8204)
Age, mean (SD) [No.], y	58 (16) [8230]	58 (16) [8204]
Age group, No./Total no. (%)		
18-29 y	459/8230 (5.6)	420/8204 (5.1)
30-39 y	721/8230 (8.8)	766/8204 (9.3)
40-49 y	1070/8230 (13.0)	1106/8204 (13.5)
50-59 y	1790/8230 (21.7)	1729/8204 (21.1)
60-69 y	1910/8230 (23.2)	1937/8204 (23.6)
70-79 y	1790/8230 (21.7)	1700/8204 (20.7)
≥80 y	490/8230 (6.0)	546/8204 (6.7)
Primary reason for admission by body system, No./Total no. (%)		
Cardiovascular	1379/7286 (18.9)	1338/7405 (18.1)
Dermatological	136/7286 (1.9)	120/7405 (1.6)
Endocrine, Metabolic, Thermoregulation and Poisoning	697/7286 (9.6)	729/7405 (9.8)
Gastrointestinal	1461/7286 (20.1)	1525/7405 (20.6)
Genito-urinary	255/7286 (3.5)	272/7405 (3.7)
Haematological/Immunological	57/7286 (0.8)	72/7405 (1.0)
Musculoskeletal	104/7286 (1.4)	105/7405 (1.4)
Neurological (including eyes)	830/7286 (11.4)	862/7405 (11.6)
Respiratory	2367/7286 (32.5)	2382/7405 (32.2)
Prior length of hospital stay, mean (SD) [No.], d	3.3 (8.1) [7293]	3.3 (7.7) [7419]
Prior duration of invasive mechanical ventilation in ICU ^a , mean (SD) [No.], h	5.3 (3.5) [8230]	5.3 (3.5) [8204]
SpO ₂ , mean (SD) [No.], %	96.0 (3.5) [8230]	95.9 (3.7) [8204]
FiO ₂ , mean (SD) [No.]	0.51 (0.21) [8230]	0.51 (0.21) [8204]
PaO ₂ , mean (SD) [No.], mm Hg	104 (48) [7638]	103 (48) [7620]
ICNARC _{H-2023} model predicted risk of death ^b , median (IQR) [No.]	0.27 (0.09-0.56) [6882]	0.26 (0.09-0.55) [7014]
APACHE II score ^c , mean (SD) [No.]	16.9 (6.5) [7317]	16.9 (6.7) [7437]

Abbreviations: APACHE II, Acute Physiology and Chronic Health Evaluation II; FiO₂, fraction of inspired oxygen; ICNARC, Intensive Care National Audit & Research Centre; ICU, intensive care unit; IQR, interquartile range; PaO₂, partial pressure of oxygen in the arterial blood; SD, standard deviation; SpO₂, peripheral oxygen saturation. SI conversion factors: To convert PaO₂ to kPa, multiply values by 0.133.

^a. Calculated since admission to critical care.

^b. ICNARC_{H-2023} was calculated using physiological measures, age, past medical history, dependency, cardiopulmonary resuscitation prior to admission, mechanical ventilation receipt, source of and primary reason for admission.³ Other than PaO₂ and FIO₂ values (last prior to randomization), physiological measures reflect information from the first 24 hours in critical care.

^c. APACHE II score (range, 0–71; higher scores indicate greater severity) was calculated using physiological measures, age and previous health status.⁴ Other than PaO₂ and FIO₂ values (last prior to randomization), physiological measures reflect information from the first 24 hours in critical care.

eTable 3. Representativeness of patients recruited to the UK-ROX trial

	Randomised patients ^a (n = 16 500)	Potentially eligible admissions across all NHS ICUs in the CMP ^b (n = 207 857)
Age, median (IQR) [No.], y	60 (48-71) [16 434]	60 (47-71) [207 857]
Sex		
Female	5652/14 805 (38.2)	79 268/207 852 (38.1)
Male	9153/14 805 (61.8)	128 584/207 852 (61.9)
Ethnic group		
Asian	506/14 805 (3.4)	11 192/207 857 (5.4)
Black	291/14 805 (2.0)	6634/207 857 (3.2)
Mixed	112/14 805 (0.8)	1873/207 857 (0.9)
White	12 279/14 805 (82.9)	161 432/207 857 (77.7)
Other ^c or not stated	1617/14 805 (10.9)	26 726/207 857 (12.9)
Index of Multiple Deprivation ^d		
1 (least deprived)	1972/14 368 (13.7)	30 956/201 843 (15.3)
2	2502/14 368 (17.4)	35 488/201 843 (17.6)
3	2953/14 368 (20.6)	39 385/201 843 (19.5)
4	3114/14 368 (21.7)	44 676/201 843 (22.1)
5 (most deprived)	3827/14 368 (26.6)	51 338/201 843 (25.4)
Body mass index, kg/m ²		
<18.5	523/14 336 (3.6)	7071/200 393 (3.5)
18.5-<25	4590/14 336 (32.0)	70 125/200 393 (35.0)
25-<30	4379/14 336 (30.5)	63 445/200 393 (31.7)
30-<40	3799/14 336 (26.5)	47 845/200 393 (23.9)
≥40	1045/14 336 (7.3)	11 907/200 393 (5.9)
Pre-existing severe respiratory disease ^e	343/14 746 (2.3)	4401/206 153 (2.1)
Recently or currently pregnant	98/4473 (2.2)	2584/63 653 (4.1)
ICNARC _{H-2023} model predicted risk of death ^f , mean (SD) [No.]	0.34 (0.29) [13 896]	0.34 (0.31) [206 957]
APACHE II score ^g , median (IQR) [No.]	16 (12-21) [14 754]	15 (11-21) [200 385]

Values are No./Total no. (%) unless otherwise indicated. Abbreviations: APACHE II, Acute Physiology and Chronic Health Evaluation II; CMP, Case Mix Programme; ICNARC, Intensive Care National Audit & Research Centre; ICU, Intensive care unit; IQR, interquartile range; NHS, National Health Service.

^a. Data to assess representativeness were not available for some patients, such as those who completely withdrew trial data or where linkage to the CMP was not possible.

^b. Potential eligibility was determined by mapping UK-ROX eligibility criteria as closely as possible to CMP data fields.

^c. Other includes those in the Chinese ethnic group, and those not in the groups otherwise listed, collected as per the National Health Service Data Dictionary definitions for 'Ethnic category'.⁵

^d. Index of Multiple Deprivation, derived from patient postcode and categorised into quintiles of the respective national distribution, measures the relative deprivation of small areas (corresponding to postcodes) according to the 2019 Indices of Multiple Deprivation for England⁶ and for Wales⁷ and the 2017 Multiple Deprivation Measure for Northern Ireland.⁸

^e. Shortness of breath with light activity due to pulmonary disease and evident within the six months prior to admission.

^f. ICNARC_{H-2023} predicted risk of in-hospital death was calculated using physiological measures, age, past medical history, dependency, cardiopulmonary resuscitation prior to admission, mechanical ventilation receipt, source of and primary reason for admission.³

^g. APACHE II score (range, 0–71; higher scores indicate greater severity) was calculated using physiological measures, age and previous health status.⁴

eTable 4. Patient baseline characteristics by data collection group

	Enhanced data collection – first 10 patients (n = 952)	Enhanced data collection – randomly selected (n = 1537)	Standard data collection (n = 13 945)
Age, median (IQR) [No.], y	60 (47-71) [952]	59 (47-71) [1537]	60 (48-71) [13 945]
Age, mean (SD) [No.], y	58 (16) [952]	58 (16) [1537]	58 (16) [13 945]
Age group			
18-29 y	45/952 (4.7)	102/1537 (6.6)	732/13945 (5.2)
30-39 y	89/952 (9.3)	131/1537 (8.5)	1267/13945 (9.1)
40-49 y	143/952 (15.0)	193/1537 (12.6)	1840/13945 (13.2)
50-59 y	193/952 (20.3)	343/1537 (22.3)	2983/13945 (21.4)
60-69 y	211/952 (22.2)	349/1537 (22.7)	3287/13945 (23.6)
70-79 y	213/952 (22.4)	321/1537 (20.9)	2956/13945 (21.2)
≥80 y	58/952 (6.1)	98/1537 (6.4)	880/13945 (6.3)
Sex			
Female	349/892 (39.1)	557/1396 (39.9)	4746/12517 (37.9)
Male	543/892 (60.9)	839/1396 (60.1)	7771/12517 (62.1)
Ethnic group			
Asian	44/892 (4.9)	43/1396 (3.1)	419/12517 (3.3)
Black	22/892 (2.5)	21/1396 (1.5)	248/12517 (2.0)
Mixed	9/892 (1.0)	12/1396 (0.9)	91/12517 (0.7)
White	736/892 (82.5)	1164/1396 (83.4)	10379/12517 (82.9)
Other ^a or not stated	81/892 (9.1)	156/1396 (11.2)	1380/12517 (11.0)
Body mass index, kg/m ²			
<18.5	30/844 (3.6)	47/1348 (3.5)	446/12144 (3.7)
18.5-<25	291/844 (34.5)	419/1348 (31.1)	3880/12144 (31.9)
25-<30	249/844 (29.5)	408/1348 (30.3)	3722/12144 (30.6)
30-<40	215/844 (25.5)	363/1348 (26.9)	3221/12144 (26.5)
≥40	59/844 (7.0)	111/1348 (8.2)	875/12144 (7.2)
Primary reason for admission by body system			
Cardiovascular	144/881 (16.3)	255/1386 (18.4)	2318/12424 (18.7)
Dermatological	15/881 (1.7)	29/1386 (2.1)	212/12424 (1.7)
Endocrine, Metabolic, Thermoregulation and Poisoning	90/881 (10.2)	134/1386 (9.7)	1202/12424 (9.7)
Gastrointestinal	168/881 (19.1)	287/1386 (20.7)	2531/12424 (20.4)
Genito-urinary	34/881 (3.9)	46/1386 (3.3)	447/12424 (3.6)
Haematological/Immunological	7/881 (0.8)	13/1386 (0.9)	109/12424 (0.9)
Musculoskeletal	17/881 (1.9)	17/1386 (1.2)	175/12424 (1.4)
Neurological (including eyes)	90/881 (10.2)	136/1386 (9.8)	1466/12424 (11.8)
Respiratory	316/881 (35.9)	469/1386 (33.8)	3964/12424 (31.9)
Pre-existing severe respiratory disease ^b	19/890 (2.1)	41/1389 (3.0)	283/12467 (2.3)
Prior length of hospital stay, median (IQR) [No.], d	1 (0-3) [882]	1 (1-3) [1389]	1 (1-3) [12 441]
Prior length of hospital stay, mean (SD) [No.], d	3.0 (6.4) [882]	3.4 (8.6) [1389]	3.3 (7.9) [12 441]
Prior duration of invasive mechanical ventilation in ICU ^c , median (IQR) [No.], h	5 (3-9) [952]	5 (2-8) [1537]	5 (2-8) [13 945]
Prior duration of invasive mechanical ventilation in ICU ^c , mean (SD) [No.], h	5.8 (3.6) [952]	5.3 (3.5) [1537]	5.3 (3.5) [13 945]

	Enhanced data collection – first 10 patients (n = 952)	Enhanced data collection – randomly selected (n = 1537)	Standard data collection (n = 13 945)
Current or suspected diagnosis ^d			
Sepsis	364/952 (38.2)	513/1537 (33.4)	4566/13 945 (32.7)
HIE	66/952 (6.9)	160/1537 (10.4)	1278/13 945 (9.2)
Acute brain injury (except HIE)	19/952 (2.0)	23/1537 (1.5)	321/13 945 (2.3)
None of the pre-specified subgroups	503/952 (52.8)	841/1537 (54.7)	7780/13 945 (55.8)
Confirmed/highly suspected COVID-19	128/952 (13.4)	82/1537 (5.3)	889/13 945 (6.4)
SpO ₂ , median (IQR) [No.], %	96 (94-99) [952]	96 (94-98) [1537]	97 (94-99) [13 945]
SpO ₂ , mean (SD) [No.], %	95.8 (4.0) [952]	95.9 (3.4) [1537]	96.0 (3.6) [13 945]
SpO ₂			
<88 %	20/952 (2.1)	25/1537 (1.6)	239/13945 (1.7)
88-92 %	134/952 (14.1)	201/1537 (13.1)	1745/13945 (12.5)
93-95 %	230/952 (24.2)	380/1537 (24.7)	3282/13945 (23.5)
>95 %	568/952 (59.7)	931/1537 (60.6)	8679/13945 (62.2)
FiO ₂ , median (IQR) [No.]	0.50 (0.35-0.70) [952]	0.45 (0.35-0.65) [1537]	0.45 (0.35-0.60) [13945]
FiO ₂ , mean (SD) [No.]	0.53 (0.23) [952]	0.51 (0.21) [1537]	0.51 (0.21) [13 945]
PaO ₂ , median (IQR) [No.], mm Hg	90 (74-116) [899]	88 (74-111) [1428]	89 (75-116) [12 931]
PaO ₂ , mean (SD) [No.], mmHg	102 (45) [899]	100 (43) [1428]	104 (49) [12 931]
PaO ₂ /FiO ₂ ratio, mm Hg			
≤100	133/899 (14.8)	210/1428 (14.7)	1526/12931 (11.8)
>100-≤200	293/899 (32.6)	486/1428 (34.0)	4520/12931 (35.0)
>200-≤300	231/899 (25.7)	360/1428 (25.2)	3364/12931 (26.0)
>300	242/899 (26.9)	372/1428 (26.1)	3521/12931 (27.2)
ICNARC _{H-2023} model predicted risk of death ^e , median (IQR) [No.]	0.26 (0.09-0.53) [843]	0.28 (0.09-0.58) [1309]	0.27 (0.09-0.55) [11744]
ICNARC _{H-2023} model predicted risk of death ^e , mean (SD) [No.]	0.33 (0.28) [843]	0.36 (0.30) [1309]	0.34 (0.29) [11744]
APACHE II score ^f , median (IQR) [No.]	16 (12-21) [890]	16 (13-21) [1391]	16 (12-21) [12 473]
APACHE II score ^f , mean (SD) [No.]	16.7 (6.5) [890]	17.1 (6.6) [1391]	16.9 (6.6) [12 473]

Values are No./Total no. (%) unless otherwise indicated.

APACHE II, Acute Physiology and Chronic Health Evaluation II; FIO₂, fraction of inspired oxygen; HIE, hypoxic-ischemic encephalopathy; ICNARC, Intensive Care National Audit & Research Centre; ICU, intensive care unit; IQR, interquartile range; PaO₂, partial pressure of oxygen in the arterial blood; SD, standard deviation; SpO₂, peripheral oxygen saturation.

SI conversion factors: To convert PaO₂ and PaO₂/FIO₂ ratio to kPa, multiply values by 0.133.

- Other includes those in the Chinese ethnic group, and those not in the groups otherwise listed, collected as per the National Health Service Data Dictionary definitions for 'Ethnic category'.⁵
- Shortness of breath with light activity due to pulmonary disease and evident within the six months prior to admission.
- Calculated since admission to critical care.
- For stratified randomisation, hierarchical classification was used to select at most one subgroup for each patient (from highest to lowest priority: HIE, sepsis, acute brain injury except HIE).
- ICNARC_{H-2023} was calculated using physiological measures, age, past medical history, dependency, cardiopulmonary resuscitation prior to admission, mechanical ventilation receipt, source of and primary reason for admission.³ Other than PaO₂ and FIO₂ values (last prior to randomisation), physiological measures reflect information from the first 24 hours in critical care.
- APACHE II score (range, 0–71; higher scores indicate greater severity) was calculated using physiological measures, age and previous health status.⁴ Other than PaO₂ and FIO₂ values (last prior to randomization), physiological measures reflect information from the first 24 hours in critical care.

eTable 5. Exposure to supplemental oxygen ^a

O ₂ exposure indicator	Conservative oxygen therapy group (n = 1252)	Usual oxygen therapy group (n = 1237)	Difference (95% CI)
Patient median FiO ₂			
Mean (SD) [No.]	0.31 (0.14) [1248]	0.35 (0.15) [1230]	-0.04 (-0.05 to -0.03)
Median (IQR) [No.]	0.25 (0.21 to 0.35) [1248]	0.30 (0.25 to 0.40) [1230]	
Patient median FiO ₂ when receiving O ₂			
Mean (SD) [No.]	0.35 (0.13) [1225]	0.37 (0.14) [1228]	-0.02 (-0.03 to -0.01)
Median (IQR) [No.]	0.30 (0.27 to 0.38) [1225]	0.31 (0.28 to 0.40) [1228]	
Patient median SpO ₂ , %			
Mean (SD) [No.]	93.3 (2.8) [1248]	95.1 (2.4) [1230]	-1.8 (-2.0 to -1.6)
Median (IQR) [No.]	93 (92 to 95) [1248]	95 (94 to 97) [1230]	
Patient median SpO ₂ when receiving O ₂ , %			
Mean (SD) [No.]	93.3 (3.0) [1225]	95.2 (2.6) [1228]	-2.0 (-2.2 to -1.8)
Median (IQR) [No.]	93 (91 to 95) [1225]	96 (94 to 97) [1228]	
Patient median SaO ₂ , %			
Mean (SD) [No.]	94.1 (2.7) [1123]	95.5 (2.7) [1091]	-1.4 (-1.6 to -1.2)
Median (IQR) [No.]	94 (92 to 96) [1123]	96 (94 to 97) [1091]	
Patient median SaO ₂ when receiving O ₂ , %			
Mean (SD) [No.]	94.0 (2.8) [1095]	95.6 (2.9) [1086]	-1.6 (-1.8 to -1.4)
Median (IQR) [No.]	94 (92 to 96) [1095]	96 (94 to 97) [1086]	
Patient median PaO ₂ , mmHg			
Mean (SD) [No.]	71.5 (13.9) [1241]	79.5 (17.9) [1227]	-8.0 (-9.3 to -6.7)
Median (IQR) [No.]	68 (64 to 76) [1241]	77 (70 to 84) [1227]	
Patient median PaO ₂ when receiving O ₂ , mmHg			
Mean (SD) [No.]	73.8 (22.8) [1217]	81.4 (19.3) [1225]	-7.6 (-9.3 to -5.9)
Median (IQR) [No.]	68 (64 to 77) [1217]	78 (71 to 86) [1225]	
Time spent on room air with SpO ₂ > 92%, h			
Mean (SD) [No.]	39.7 (55.1) [1234]	26.1 (45.1) [1216]	13.6 (9.6 to 17.6)
Median (IQR) [No.]	14 (0 to 59) [1234]	3 (0 to 32) [1216]	
Time spent receiving O ₂ when SpO ₂ > 92%, h			
Mean (SD) [No.]	41.4 (45.6) [1234]	95.4 (72.9) [1216]	-54.0 (-58.8 to -49.2)
Median (IQR) [No.]	24 (8 to 59) [1234]	79 (30 to 155) [1216]	
Time spent when SpO ₂ is 88-92% ^b , h			
Mean (SD) [No.]	62.6 (62.3) [1234]	27.2 (39.1) [1216]	35.5 (31.4 to 39.6)
Median (IQR) [No.]	40 (10 to 100) [1234]	10 (2 to 35) [1216]	
Time spent when SpO ₂ < 88%, h			
Mean (SD) [No.]	3.2 (6.5) [1234]	2.3 (7.3) [1216]	0.9 (0.4 to 1.4)
Median (IQR) [No.]	1 (0 to 4) [1234]	0 (0 to 2) [1216]	
Total O ₂ exposure, 100%-equivalent h ^c			
Mean (SD) [No.]	20.3 (27.4) [1234]	28.7 (31.9) [1216]	-8.4 (-10.8 to -6.0)
% difference		[reference]	-29.3 (-37.6 to -20.9)
Median (IQR) [No.]	8 (2 to 28) [1234]	16 (6 to 40) [1216]	

Abbreviations: CI, confidence interval; FiO₂, fraction of inspired oxygen; IQR, interquartile range; O₂, (supplemental) oxygen; PaO₂, partial pressure of oxygen in the arterial blood; SaO₂, arterial hemoglobin oxygen saturation; SD, standard deviation; SpO₂, peripheral oxygen saturation.

^a. Data were available for the subset of patients for whom enhanced data collection was conducted and were collected from the day of randomization to day 10 post-randomization, intensive care unit discharge, death or refusal of consent/opinion (whichever came first).

^b. Target range of conservative oxygen therapy.

^c. Cumulative exposure to supplemental O₂ (whereby time spent on room air represents zero supplemental O₂, and 1 h on 100% O₂ or 2 h on 60.5% O₂ would both represent 1 100%-equivalent h of O₂).

eTable 6. Exposure to supplemental oxygen by data collection group ^a

O ₂ exposure indicator	First 10 patients (n = 952)			Random sample patients (n = 1537)		
	Conservative oxygen therapy group (n = 486)	Usual oxygen therapy group (n = 466)	Difference (95% CI)	Conservative oxygen therapy group (n = 766)	Usual oxygen therapy group (n = 771)	Difference (95% CI)
Patient median SpO ₂						
Mean (SD) [No.]	93.1 (3.1) [484]	95.2 (2.6) [462]	-2.1 (-2.5 to -1.7)	93.5 (2.7) [764]	95.0 (2.4) [768]	-1.6 (-1.9 to -1.3)
Median (IQR) [No.]	93 (91 to 95) [484]	96 (94 to 97) [462]		93 (92 to 95) [764]	95 (94 to 96) [768]	
Patient median SpO ₂ when receiving O ₂						
Mean (SD) [No.]	93.0 (3.2) [468]	95.3 (2.7) [461]	-2.3 (-2.7 to -1.9)	93.4 (2.9) [757]	95.2 (2.5) [767]	-1.7 (-2.0 to -1.4)
Median (IQR) [No.]	92 (91 to 95) [468]	96 (94 to 97) [461]		93 (92 to 95) [757]	95 (94 to 97) [767]	
Patient median FiO ₂ , %						
Mean (SD) [No.]	0.32 (0.15) [484]	0.35 (0.15) [462]	-0.04 (-0.06 to -0.02)	0.30 (0.13) [764]	0.35 (0.15) [768]	-0.05 (-0.06 to -0.04)
Median (IQR) [No.]	0.25 (0.21 to 0.35) [484]	0.30 (0.25 to 0.40) [462]		0.25 (0.21 to 0.35) [764]	0.30 (0.25 to 0.40) [768]	
Patient median FiO ₂ when receiving O ₂ , %						
Mean (SD) [No.]	0.36 (0.14) [468]	0.37 (0.14) [461]	-0.02 (-0.04 to -0.00)	0.34 (0.12) [757]	0.37 (0.14) [767]	-0.03 (-0.04 to -0.02)
Median (IQR) [No.]	0.30 (0.26 to 0.40) [468]	0.31 (0.28 to 0.40) [461]		0.30 (0.27 to 0.35) [757]	0.31 (0.28 to 0.40) [767]	
Patient median SaO ₂ , %						
Mean (SD) [No.]	94.0 (2.9) [431]	95.6 (2.7) [408]	-1.6 (-2.0 to -1.2)	94.1 (2.6) [692]	95.4 (2.7) [683]	-1.3 (-1.6 to -1.0)
Median (IQR) [No.]	94 (92 to 96) [431]	96 (94 to 97) [408]		94 (93 to 96) [692]	96 (94 to 97) [683]	
Patient median SaO ₂ when receiving O ₂ , %						
Mean (SD) [No.]	93.9 (2.8) [414]	95.6 (3.2) [405]	-1.7 (-2.1 to -1.3)	94.0 (2.7) [681]	95.6 (2.7) [681]	-1.6 (-1.9 to -1.3)
Median (IQR) [No.]	94 (92 to 96) [414]	96 (95 to 98) [405]		94 (92 to 96) [681]	96 (94 to 97) [681]	
Patient median PaO ₂ , mmHg						
Mean (SD) [No.]	71.1 (15.1) [479]	79.8 (17.1) [461]	-8.7 (-10.8 to -6.6)	71.7 (13.0) [762]	79.2 (18.4) [766]	-7.5 (-9.1 to -5.9)
Median (IQR) [No.]	68 (63 to 75) [479]	77 (70 to 86) [461]		68 (65 to 76) [762]	77 (70 to 83) [766]	
Patient median PaO ₂ when receiving O ₂ , mmHg						
Mean (SD) [No.]	73.8 (23.9) [462]	82.0 (19.3) [460]	-8.3 (-11.1 to -5.5)	73.9 (22.1) [755]	81.0 (19.3) [765]	-7.2 (-9.3 to -5.1)
Median (IQR) [No.]	68 (63 to 76) [462]	79 (71 to 89) [460]		69 (64 to 77) [755]	77 (71 to 86) [765]	
Time spent on room air with SpO ₂ > 92%, h						
Mean (SD) [No.]	37.3 (55.0) [480]	26.0 (45.8) [456]	11.4 (4.9 to 17.9)	41.2 (55.2) [754]	26.1 (44.7) [760]	15.1 (10.0 to 20.2)

O ₂ exposure indicator	First 10 patients (n = 952)			Random sample patients (n = 1537)		
	Conservative oxygen therapy group (n = 486)	Usual oxygen therapy group (n = 466)	Difference (95% CI)	Conservative oxygen therapy group (n = 766)	Usual oxygen therapy group (n = 771)	Difference (95% CI)
Median (IQR) [No.]	12 (0 to 52) [480]	2 (0 to 34) [456]		16 (0 to 62) [754]	4 (0 to 31) [760]	
Time spent receiving O ₂ when SpO ₂ > 92%, h						
Mean (SD) [No.]	37.3 (45.9) [480]	97.0 (75.2) [456]	-59.8 (-67.8 to -51.8)	44.0 (45.2) [754]	94.4 (71.5) [760]	-50.4 (-56.4 to -44.4)
Median (IQR) [No.]	20 (6 to 49) [480]	82 (28 to 150) [456]		28 (10 to 64) [754]	78 (31 to 156) [760]	
Time spent when SpO ₂ is 88-92% ^b , h						
Mean (SD) [No.]	65.9 (66.2) [480]	28.2 (42.5) [456]	37.7 (30.6 to 44.8)	60.5 (59.7) [754]	26.5 (36.8) [760]	34.0 (29.0 to 39.0)
Median (IQR) [No.]	39 (12 to 104) [480]	9 (2 to 34) [456]		41 (9 to 97) [754]	11 (2 to 35) [760]	
Time spent when SpO ₂ < 88%, h						
Mean (SD) [No.]	3.5 (6.6) [480]	2.4 (7.4) [456]	1.1 (0.2 to 2.0)	3.1 (6.4) [754]	2.3 (7.3) [760]	0.8 (0.1 to 1.5)
Median (IQR) [No.]	1 (0 to 4) [480]	0 (0 to 1) [456]		1 (0 to 4) [754]	0 (0 to 2) [760]	
Total O ₂ exposure, 100%-equivalent h ^c						
Mean (SD) [No.]	22.0 (30.4) [480]	31.2 (35.2) [456]	-9.2 (-13.4 to -5.0)	19.3 (25.2) [754]	27.2 (29.7) [760]	-7.9 (-10.7 to -5.1)
% difference		[reference]	-29.5 (-42.9 to -16.0)		[reference]	-29.0 (-39.3 to -18.8)
Median (IQR) [No.]	8 (2 to 31) [480]	19 (5 to 44) [456]		9 (2 to 26) [754]	15 (6 to 39) [760]	

Abbreviations: CI, confidence interval; FiO₂, fraction of inspired oxygen; IQR, interquartile range; O₂, (supplemental) oxygen; PaO₂, partial pressure of oxygen in the arterial blood; SaO₂, arterial hemoglobin oxygen saturation; SD, standard deviation; SpO₂, peripheral oxygen saturation.

^a. Data were available for the subset of patients for whom enhanced data collection was conducted and were collected from the day of randomization to day 10 post-randomization, intensive care unit discharge, death or refusal of consent/opinion (whichever came first).

^b. Target range of conservative oxygen therapy.

^c. Cumulative exposure to supplemental O₂ (whereby time spent on room air represents zero supplemental O₂, and 1 h on 100% O₂ or 2 h on 60.5% O₂ would both represent 1 100%-equivalent h of O₂).

eTable 7. Missing data in analysis variables

	Missing values	Imputation model	Auxiliary variables ^a
Baseline variables, No. (%)			
Allocated treatment group	0 (0.0)	None required	N/A
Age	0 (0.0)	None required	N/A
Current or suspected diagnosis ^b	0 (0.0)	None required	N/A
Confirmed/highly suspected COVID-19	0 (0.0)	None required	N/A
SpO ₂ at randomization	0 (0.0)	None required	N/A
PaO ₂ /FiO ₂ ratio at randomization	1176 (7.2)	Single ^c	N/A
Date of randomization	0 (0.0)	None required	N/A
Ethnic group	1630 (9.9)	None ^d	N/A
Primary outcome, No. (%)			
90-d mortality	40 (0.2)	Logit	Duration of acute hospital stay
Secondary outcomes, No. (%)			
Duration of ICU stay	1653 (10.1)	Censoring ^e	N/A
Duration of acute hospital stay	1677 (10.2)	Censoring ^e	N/A
Days alive and free from organ support at 30 d	1639 (10.0)	Ordered logit	90-d mortality; mortality at ICU discharge
Mortality at ICU discharge	1647 (10.0)	Logit	90-d mortality
Mortality at acute hospital discharge	1641 (10.0)	Logit	90-d mortality; duration of acute hospital stay
60-d mortality	1412 (8.6)	Logit	90-d mortality; 1-y mortality
1-y mortality, No (%) [No. reached 1 y]	1661 (12.7) [13052]	Logit	None

Abbreviations: HIE, hypoxic-ischemic encephalopathy; FiO₂, fraction of inspired oxygen; ICU, intensive care unit; PaO₂, partial pressure of oxygen in the arterial blood; SpO₂, peripheral oxygen saturation.

^a. Allocated treatment group, age, current or suspected diagnosis, confirmed/highly suspected COVID-19, SpO₂ at randomization, PaO₂/FiO₂ ratio at randomization and date of randomization were included in all imputation models.

^b. Hierarchical subgroup upon which randomization was stratified: hypoxic ischemic encephalopathy (HIE); sepsis; acute brain injury (except HIE); none of the above.

^c. Missing values of PaO₂/FiO₂ ratio were singly imputed from SpO₂/FiO₂ ratio using the formula of Brown et al 2016.⁹

^d. Ethnic group was only included in the analysis model for subgroup analysis by ethnic group and only patients with complete data for ethnic group were included.

^e. Patients missing duration of ICU stay or duration of acute hospital stay were included in the model and censored at 1 hour (for duration of ICU stay) or 1 day (for duration of acute hospital stay) following randomization.

eTable 8. Additional summary statistics for continuous secondary clinical outcomes

Outcome	Conservative oxygen therapy group	Usual oxygen therapy group
Duration of ICU stay, Mean (SD) [No.], d		
Overall	11.0 (13.0) [7333]	11.4 (13.8) [7448]
ICU survivors	12.3 (14.1) [5211]	12.8 (14.9) [5290]
ICU non-survivors	7.8 (9.3) [2122]	7.8 (9.9) [2158]
Duration of acute hospital stay, Mean (SD) [No.], d		
Overall	22.9 (23.1) [7323]	23.1 (23.3) [7434]
Hospital survivors	29.1 (24.9) [4791]	29.5 (25.0) [4906]
Hospital non-survivors	11.3 (12.7) [2532]	10.9 (12.9) [2528]
Days alive and free from organ support at 30 d, Mean (SD) [No.] ^a	12.9 (12.2) [7327]	12.8 (12.1) [7444]
Days free from organ support at 30 d among survivors, Mean (SD) [No.]	19.6 (9.1) [4933]	19.3 (9.1) [5054]
Days free from respiratory support at 30 d among survivors		
Median (IQR) [No.]	25 (18 to 27) [4933]	24 (17 to 27) [5054]
Mean (SD) [No.]	20.9 (9.1) [4933]	20.5 (9.1) [5054]
Days free from cardiovascular support at 30 d among survivors		
Median (IQR) [No.]	23 (16 to 27) [4933]	23 (16 to 26) [5054]
Mean (SD) [No.]	20.1 (8.6) [4933]	19.8 (8.8) [5054]
Days free from renal support at 30 d among survivors		
Median (IQR) [No.]	30 (30 to 30) [4933]	30 (30 to 30) [5054]
Mean (SD) [No.]	28.7 (4.3) [4933]	28.5 (4.7) [5054]

Abbreviations: ICU, intensive care unit; IQR, interquartile range; SD, standard deviation.

^a Ordinal composite outcome with patients who died on or before day 30 assigned the worst possible score of –1. Surviving patients were ranked according to the number of calendar days on which any respiratory, cardiovascular or renal support was received at any time during that day, starting from day 1 (the day of randomization) up to and including day 30.

eTable 9. Serious adverse events

	Conservative oxygen therapy group (n = 8230)		Usual oxygen therapy group (n = 8204)	
	Events, No.	Patients, No. (%)	Events, No.	Patients, No. (%)
Any serious adverse event		58 (0.7)		29 (0.4)
Specified serious adverse events				
Sinus tachycardia	2	2 (<0.1)	1	1 (<0.1)
Supraventricular tachycardia	12	12 (0.1)	8	8 (0.1)
Atrial fibrillation	16	14 (0.2)	15	13 (0.2)
Myocardial ischemia/infarction	19	19 (0.2)	11	11 (0.1)
Mesenteric ischemia	7	7 (0.1)	0	0 (0.0)
Other serious adverse events				
Bradycardia	1	1 (<0.1)	0	0 (0.0)
Bradycardic and asystole	1	1 (<0.1)	0	0 (0.0)
Cardiac arrest	1	1 (<0.1)	0	0 (0.0)
Cardiovascular instability	1	1 (<0.1)	0	0 (0.0)
Ischemic changes to brain from CT	1	1 (<0.1)	0	0 (0.0)
Multiple segmental cerebral infarcts	1	1 (<0.1)	0	0 (0.0)
Pneumothorax	0	0 (0.0)	2	1 (<0.1)
Respiratory arrest	1	1 (<0.1)	0	0 (0.0)
Severe hypoxemia	1	1 (<0.1)	0	0 (0.0)

Recorded from the time of randomization to intensive care unit discharge or 90 d (whichever came first).
Abbreviations: CT, computed tomography.

eTable 10. Comparison of achieved oxygenation across different trials in critically ill patients

Trial characteristics	UK-ROX	HOT-COVID ¹⁰	ICONIC ¹¹	PILOT ¹²	HOT-ICU ¹³	O₂-ICU ¹⁴	ICU-ROX ¹⁵
Year published	2025	2024	2023	2022	202	2021	2020
Recruitment dates	05/2021 to 11/2024	08/2020 to 03/2023	11/2018 to 11/2021	07/2018 to 08/2021*	06/2017 to 08/2020	02/2015 to 10/2018	09/2015 to 05/2018
Number of participants:							
Total with data	16,434	726	664	2541	2910	400	965
Mechanically ventilated	16,434	172	664	2541	1704	295	965
Oxygenation target:							
Mode	SpO ₂	PaO ₂	PaO ₂ / SpO ₂	SpO ₂	PaO ₂	PaO ₂	SpO ₂
Lower group SpO ₂ target	90% (88-92%)	Approx 90%	91-94%	90%	Approx 90%	Approx 90-97%	91-96%
Lower group PaO ₂ target (mmHg)	Approx 60	60	55-80	60	60	60-90	60
Higher group SpO ₂ target	Usual care	Approx 97%	96-100	98%	Approx 97%	Approx 98-100%	91-100%
Higher group PaO ₂ target (mmHg)	Usual care	90	110-150	110	90	105-135	Not reported
Achieved oxygenation ^a							
Lower group SpO ₂	93.3%	93%	95%	94%	93%	96%	Not reported
Higher group SpO ₂	95.2%	96%	99%	97%	96%	97%	Not reported
SpO ₂ difference (high-low)	2%	3%	4%	3%	3%	1%	/
Lower group PaO ₂	71.5	71	75	Not reported	71	81	82 (estimated)
Higher group PaO ₂	79.5	91	115	Not reported	93	96	94 (estimated)
PaO ₂ difference (high-low)	8.0	20	40	Not reported	23	14	12 (estimated)
Lower group FiO ₂	0.31	0.54	0.36	0.31	0.43	0.4	0.30 (estimated)
Higher group FiO ₂	0.35	0.68	0.48	0.45	0.56	0.51	0.35 (estimated)
FiO ₂ difference (high-low)	0.04	0.14	0.12	0.15	0.13	0.09	0.05 (estimated)

Abbreviations: SpO₂, peripheral oxygen saturation; PaO₂, partial pressure of oxygen in the arterial blood; FiO₂, fraction of inspired oxygen;

^a Method of reporting inconsistent between median, means and time-weighted mean values. Data may also have been collected differently across trial. Some values estimated from information provided in manuscripts when precise values not given.

eReferences

1. Harrison DA, Brady AR, Rowan K. Case mix, outcome and length of stay for admissions to adult, general critical care units in England, Wales and Northern Ireland: the Intensive Care National Audit & Research Centre Case Mix Programme Database. *Crit Care*. Apr 2004;8(2):R99-111. doi:10.1186/cc2834
2. NHS England. Civil Registrations of Death. <https://digital.nhs.uk/services/data-access-request-service-dars/dars-products-and-services/data-set-catalogue/civil-registrations-of-death>
3. ICNARC. The ICNARCH-2023 model. Accessed April 9th, 2025. <https://www.icnarc.org/wp-content/uploads/2024/05/The-ICNARC-Model-2023.pdf.pdf>
4. Knaus WA, Draper EA, Wagner DP, Zimmerman JE. APACHE II: a severity of disease classification system. *Crit Care Med*. Oct 1985;13(10):818-29.
5. NHS Data Model and Dictionary: Ethnic Category. Accessed 19 May 2025, 2025. https://www.datadictionary.nhs.uk/data_elements/ethnic_category.html
6. UK Government. English indices of deprivation 2019. Accessed April 9th, 2025. <https://www.gov.uk/government/statistics/english-indices-of-deprivation-2019>
7. Welsh Government. Welsh Index of Multiple Deprivation. Accessed April 9th, 2025.
8. Northern Ireland Statistics and Research Agency. Northern Ireland Multiple Deprivation Measure. Accessed April 9th, 2025. <https://www.nisra.gov.uk/statistics/deprivation/northern-ireland-multiple-deprivation-measure-2017-nimdm2017>
9. Brown SM, Grissom CK, Moss M, et al. Nonlinear Imputation of Pao₂/Fio₂ From Spo₂/Fio₂ Among Patients With Acute Respiratory Distress Syndrome. *Chest*. Aug 2016;150(2):307-13. doi:10.1016/j.chest.2016.01.003
10. Nielsen FM, Klitgaard TL, Siegemund M, et al. Lower vs Higher Oxygenation Target and Days Alive Without Life Support in COVID-19: The HOT-COVID Randomized Clinical Trial. *JAMA*. Apr 09 2024;331(14):1185-1194. doi:10.1001/jama.2024.2934
11. van der Wal LI, Grim CCA, Del Prado MR, et al. Conservative versus Liberal Oxygenation Targets in Intensive Care Unit Patients (ICONIC): A Randomized Clinical Trial. *Am J Respir Crit Care Med*. Oct 01 2023;208(7):770-779. doi:10.1164/rccm.202303-0560OC
12. Semler MW, Casey JD, Lloyd BD, et al. Oxygen-Saturation Targets for Critically Ill Adults Receiving Mechanical Ventilation. *N Engl J Med*. Nov 10 2022;387(19):1759-1769. doi:10.1056/NEJMoa2208415

13. Schjørring OL, Klitgaard TL, Perner A, et al. Lower or Higher Oxygenation Targets for Acute Hypoxemic Respiratory Failure. *N Engl J Med*. Apr 08 2021;384(14):1301-1311. doi:10.1056/NEJMoa2032510
14. Gelissen H, de Grooth HJ, Smulders Y, et al. Effect of Low-Normal vs High-Normal Oxygenation Targets on Organ Dysfunction in Critically Ill Patients: A Randomized Clinical Trial. *JAMA*. Sep 14 2021;326(10):940-948. doi:10.1001/jama.2021.13011
15. Mackle D, Bellomo R, Bailey M, et al. Conservative Oxygen Therapy during Mechanical Ventilation in the ICU. *N Engl J Med*. Mar 12 2020;382(11):989-998. doi:10.1056/NEJMoa1903297